CAS problem-1 DATE:- 27-01-2026

The matrices A and B are given below:

A = $\begin{bmatrix} 1 & 2 \\ 2 & -2 \end{bmatrix}$ and B = $\begin{bmatrix} 1 & -2 & 0 \\ -2 & 5 & 0 \\ 0 & 0 & 2 \end{bmatrix}$

Find the eigen values and the associated normalized eigen vectors of the above matrices using R software.

Solution:-

A<- matrix (c(1, 2, 2, -2) , nrow = 2 ,ncol = 2,byrow = TRUE)

A

OUTPUT:-

```
    [,1] [,2]
[1,]    1    2
[2,]    2   -2
```

B<- matrix (c(1, -2 ,0, -2, 5, 0 ,0, 0, 2),nrow = 3,ncol=3)

B

OUTPUT:-

```
[,1] [,2] [,3]
[1,]    1   -2    0
[2,]   -2    5    0
[3,]    0    0    2
```

eigen(A)

OUTPUT:-

```
eigen() decomposition
$values
[1]  2 -3

$vectors
           [,1]       [,2]
[1,] -0.8944272 -0.4472136
[2,] -0.4472136  0.8944272
```

eigen(B)

OUTPUT:-

```
eigen() decomposition
$values
[1] 5.8284271 2.0000000 0.1715729

$vectors
           [,1] [,2]      [,3]
[1,] -0.3826834    0 0.9238795
[2,]  0.9238795    0 0.3826834
[3,]  0.0000000    1 0.0000000
```

CONCLUSION:-

Matrix A has two eigenvalues 2 and -3 , with corresponding eigenvectors. This shows how matrix A scales vector in specific directions. And matrix B has three eigenvalues 5.8284271 , 2 and 0.1715729 with corresponding eigenvectors. This shows that matrix B scales vectors differently along three specific directions.